



## Four clinical ECG sources

PTB-XL | Georgia | Chapman | Ningbo  
77,333 records -> 66,379 retained, 13 harmonised labels

10 s extraction

### Native 500 Hz signal

quality indicators measured  
before decimation

quality  
indicators

### Anti-aliased resampling to 100 Hz

model input: 12 x 1000

RQ1

### External generalisation

leave-one-source-out, six compact encoders  
macro AUPRC across unseen clinical sources

held-out predictions

RQ2

### Reliability under source and quality shift

- a what shifts between the sources  
six indicators, four sources
- ↓
- b do those shifts matter diagnostically?  
within-source strata, prevalence-matched dAUPRC
- ↓
- c can the model tell when it is unreliable?  
E-AURC, and whether it degrades with the signal

externally evaluated encoders

RQ3

### Deployment-preserved performance

int8 post-training quantisation | STM32F411 at 100 MHz  
Flash | SRAM | measured latency | accuracy-resource trade-off

Generalises? -> Can we trust it? -> Can we deploy it?